

PENGUINS

With their two-toned plumage, erect posture, and waddling walk, penguins are instantly recognizable as the classic symbol of the southern oceans.

Residents of the coastal regions of the southern hemisphere, all penguins are adapted to life in cold water. Most species live around islands encircling Antarctica, although some are found on southern coastlines of South America, Africa, and Australasia.

These flightless birds of the order Sphenisciformes probably descended from a common ancestor shared with the albatrosses. They may also be distant cousins of the divers from the northern hemisphere.

SPECIAL ADAPTATIONS

Penguins have legs set far back towards the tail, a feature that provides excellent propulsion in water and is seen also in birds such as divers and grebes. On land penguins walk upright, but with webbed feet placed flat on the ground their gait is ungainly. In common with other flightless birds, penguins have reduced wings, but these are modified in such a way that they can be used as flippers. In effect, penguins “fly” under water.

A penguin's short, densely packed feathers have downy bases that trap warm air, and a layer of fat beneath the skin provides further insulation. The feather tips are greasy and waterproof, lubricated by oil produced from a large gland on the rump. A complex system of blood-flow through the legs and feet ensures that the penguin's body is not chilled by standing on snow or ice. The largest penguin species incubate their eggs on their feet. All penguins have counter-shaded plumage (dark above and pale below), which in the sea provides camouflage against ocean-going predators, such as leopard seals.

FORAGING AND NESTING

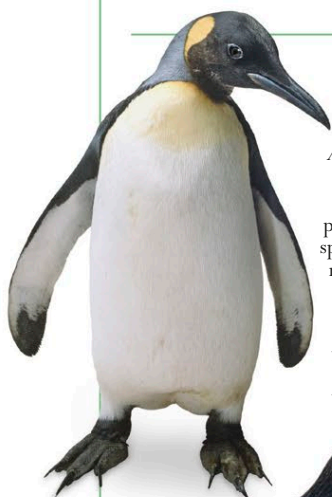
Penguins can dive for fish, shrimp, and krill more than 200 times each day. When they have eggs to incubate and young to tend, the parents take it in turns to forage. In the case of emperor penguins, one of the few species to breed on the Antarctic pack ice, the males incubate the eggs alone throughout the winter while the females remain feeding at sea. Most penguins are colonial breeders and often

return to the same nest site each season.

PHYLUM	CHORDATA
CLASS	AVES
ORDER	SPHENISCIFORMES
FAMILIES	1
SPECIES	18

DEBATE FLYING ANCESTORS?

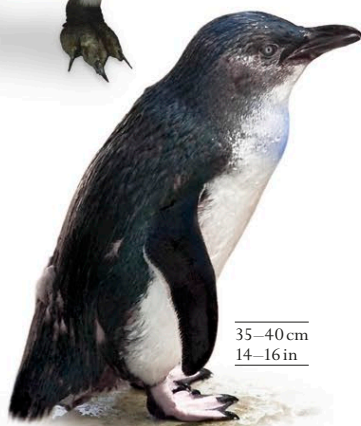
In the early 20th century, flightless birds were popularly believed to be primitive. It was hoped that study of their embryos would reveal evidence of a direct link to a dinosaurian past. Penguin eggs became sought after, and members of Robert Falcon Scott's last expedition (1910–13) made a courageous journey to an emperor rookery in the bleak Antarctic winter. By the time the eggs they collected received scientific attention, the embryo theory had been disproved. Modern studies of anatomy, fossils, and DNA show that the ancestors of penguins, and other flightless birds, could fly.



KING PENGUIN
Aptenodytes patagonicus
F: Spheniscidae

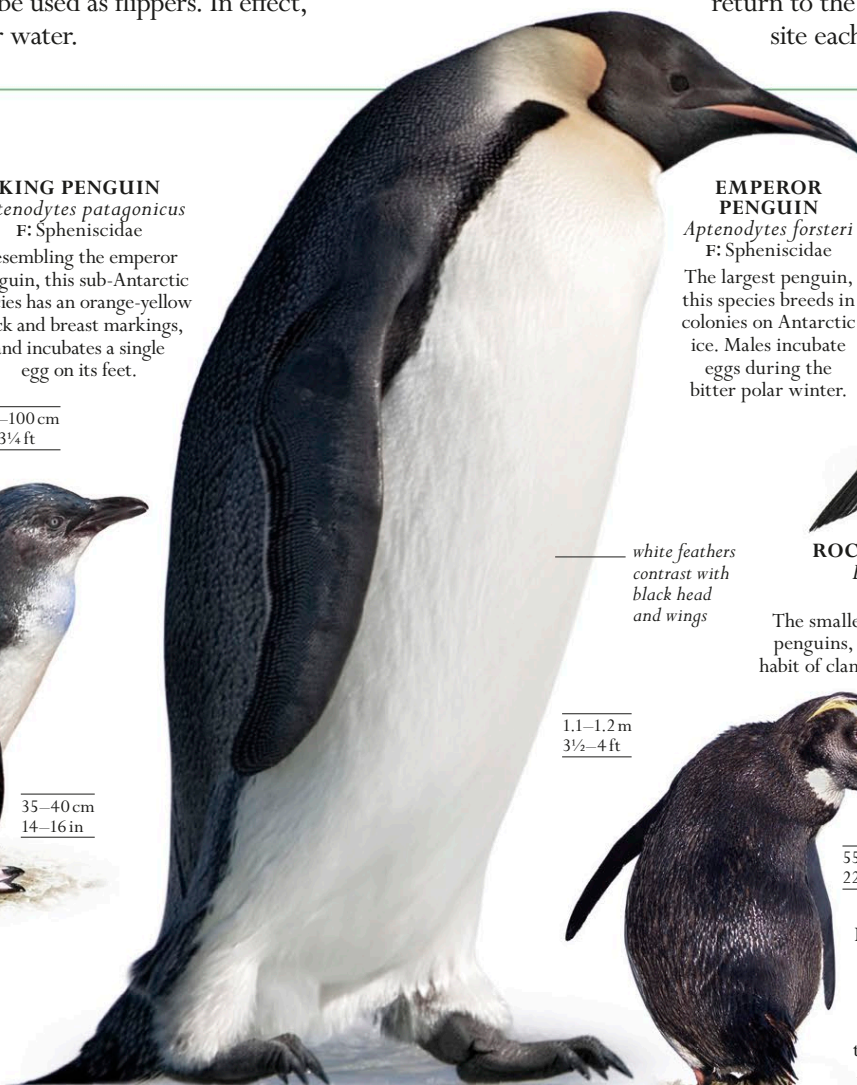
Resembling the emperor penguin, this sub-Antarctic species has an orange-yellow neck and breast markings, and incubates a single egg on its feet.

90–100 cm
3–3¼ ft



LITTLE PENGUIN
Eudyptula minor
F: Spheniscidae

A burrow-nester, this bird is the smallest of all penguins. It occurs along the coasts of S. Australia and New Zealand.



EMPEROR PENGUIN
Aptenodytes forsteri
F: Spheniscidae

The largest penguin, this species breeds in colonies on Antarctic ice. Males incubate eggs during the bitter polar winter.

white feathers
contrast with
black head
and wings

1.1–1.2 m
3½–4 ft



ROCKHOPPER PENGUIN
Eudyptes chrysocome
F: Spheniscidae

The smallest of the sub-Antarctic crested penguins, this bird gets its name from its habit of clambering over rocks and boulders.

45–58 cm
18–23 in

yellow
plume



FIORDLAND PENGUIN
Eudyptes pachyrhynchus
F: Spheniscidae

Nesting in the cool coastal forests of S. New Zealand, this species has hair-like crest feathers and a red bill typical of *Eudyptes* penguins.

55–60 cm
22–23½ in



MACARONI PENGUIN
Eudyptes chrysolophus
F: Spheniscidae

This bird inhabits islands in the far southern Atlantic and Indian oceans but is the only crested penguin to breed on the Antarctic Peninsula.

70 cm
28 in